

BIOSZEN User Manual (English)

A practical guide to run BIOSZEN from raw files to reproducible outputs.

The screenshot shows the BIOSZEN Graphics and Analysis Center interface. At the top, there is a header with the BIOSZEN logo, the text "Graphics and Analysis Center", and three buttons: "Language", "Light Theme", and "Dark Theme". Below the header, there are two tabs: "Plots and Stats" and "Growth Parameter Extraction". The "Plots and Stats" tab is active. On the left side of the main content area, there is a sidebar with several sections: "Load Data (.xlsx/.xls/.csv)" with a "Browse..." button and "No file selected" text; "Load Curves (.xlsx/.xls/.csv)" with a "Browse..." button and "No file selected" text; "Load design metadata (.xlsx)" with a "Browse..." button and "No file selected" text; "Instructions (download)" button; "Reference input files (download)" button; "Plot setup" section with "Scope:" (radio buttons for "By Strain" and "Combined", with "By Strain" selected), "Strain:" dropdown menu, "Plot:" (radio buttons for "Curves" and "Normal distribution", with "Curves" selected), and a checkbox for "Normal distribution". On the right side of the main content area, there is a large empty space for the plot. A "Quick options" button is located in the top right corner of the main content area.

IMPORTANT: If possible, use **Platemap + Curves** mode. It gives the best support for statistics, replicate QC, and complete exports.

TIP: Keep this manual open while working. Each section includes both quick actions and deeper technical reference.

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1. Before You Start

Requirements:

- R $\geq$ 4.1.
- BIOSZEN launched from `app.R` or `BIOSZEN::run_app()`.
- Data file for **Load Data** in Excel (`.xlsx` , `.xls`) or CSV (`.csv`).
- Curves file for **Load Curves** in Excel (`.xlsx` , `.xls`) or CSV (`.csv`) when curves are not embedded in the main workbook.

Reference templates available in-app (**Reference input files (download)**) and in:

- `inst/app/www/reference_files/`

Template files:

- [Ejemplo_platemap_parametros.xlsx](#)
- [Ejemplo_curvas.xlsx](#)
- [Ejemplo_parametros_agrupados.xlsx](#)
- [Ejemplo_input_summary_mean_sd.xlsx](#)

NOTE: First launch may install packages into local `R_libs`. Keep that folder to avoid reinstalling dependencies.

2. Fast Start by Scenario

Scenario A: I have raw plate data and curves (recommended)

1. Load platemap in **Load Data**.
2. Load curves file in **Load Curves**.
3. Select scope and plot type.
4. Apply filters and replicate QC.
5. Run stats and annotations.
6. Export plot, tables, metadata, and ZIP bundle.

Scenario B: I only have grouped or summary data

1. Load grouped/summary workbook in **Load Data**.

2. Configure plots and filters.
3. Run available stats for that mode.
4. Export plots and metadata.

Scenario C: I need performance on larger datasets

1. Start with `.csv` in **Load Data**.
2. Keep selected parameters small while iterating.
3. Add overlays/advanced layers only near final export.

Plot setup
Scope:
☒ By Strain
☐ Combined
Strain:

Cepa1

Parameter:

Parameter 1

Plot:
☒ Boxplot
☐ Barplot
☐ Violin
☐ Curves
☐ Stacked
☐ Correlation
☐ Heatmap
☐ Correlation Matrix
☐ Normalize by control

3. Choose an Input Mode

- **Platemap + Curves**
Best when: You want full workflow depth.
Main limitations: Requires strict well mapping and sheet structure.
- **Grouped parameters**
Best when: Parameter-only analysis.
Main limitations: Curves require embedded `Curves_Summary` -type sheets (or a separate file in **Load Curves**).
- **Summary (Mean/SD/N)**
Best when: Raw replicate rows are unavailable.
Main limitations: Some normality/non-parametric routes may be limited.
- **CSV mode**
Best when: High-volume data and faster IO.
Main limitations: Metadata roundtrip still uses `.xlsx`.

4. Input Specifications

4.1 Platemap workbook

Required sheets:

- `Datos` : metadata + parameters.
- `PlotSettings` : default axis settings by parameter.

`Datos` expected columns:

- `Well` : well ID (`A1` , `B3` , etc.), required for curves linking.
- `Strain` : strain or biological group.
- `Media` : condition/treatment (`Control` , `Drug A` , etc.).
- `BiologicalReplicate` : biological replicate ID (`1` , `2` , `3` , ...).
- `TechnicalReplicate` : technical replicate within each biological replicate (`A` , `B` , `C` or `1` , `2` , `3`).
- `Replicate` (compatibility): legacy alternate biological replicate field.
- `Orden` : integer used for plotting/export ordering.
- Parameter columns: one or more numeric analytes/metrics.

Practical consistency rule:

- `Strain` + `Media` + `BiologicalReplicate` + `TechnicalReplicate` should identify each experimental row consistently.

`PlotSettings` expected columns:

- `Parameter`
- `Y_Max`
- `Interval`
- `Y_Title`

4.2 Curves file

Excel (`.xlsx` , `.xls`):

- `Sheet1` : first column `Time` , remaining columns by well (`A1` , `A2` , ...).
- `Sheet2` : `X_Max` , `Interval_X` , `Y_Max` , `Interval_Y` , `X_Title` , `Y_Title` .

CSV (`.csv`):

- First column `Time` , remaining columns by well (`A1` , `A2` , ...).

- Axis settings are auto-generated:
 - `X_Max` and `Y_Max` : observed maxima.
 - `Interval_X` and `Interval_Y` : `max/4` .
 - `X_Title` and `Y_Title` : blank by default.

WARNING: Curves merge failures are usually caused by inconsistent well names between platemap and curves (`Well` vs curve headers).

4.3 Grouped parameters mode

- Load grouped workbook in **Load Data**.
- Designed for parameter plots/statistics from grouped sheets (for example `Parametro_1` , `Parametro_2` , ...).
- Optional embedded curves are supported via summary-curve sheets in the same workbook.
- Keep using **Load Data** for grouped workbooks (do not upload grouped files in **Load Curves**).

4.4 Summary mode

- Load summary workbook in **Load Data**.
- BIOSZEN detects parameter summaries from any of these sheet names:
 - `Parameters_Summary`
 - `Parametros_Summary`
 - `Summary_Parameters`
 - `Resumen_Parametros`
- BIOSZEN detects embedded curve summaries from any of these sheet names:
 - `Curves_Summary`
 - `Curvas_Summary`
 - `Summary_Curves`
 - `Resumen_Curvas`
- Useful when row-level raw replicates are unavailable.
- Curves plots require either a valid **Load Curves** file or an embedded curve-summary sheet.

4.5 CSV mode

- **Load Data** accepts `.csv` and auto-detects delimiter (`,` , `;` , tab, `|`).
- BIOSZEN attempts profile conversion when CSV is not already platemap-ready.
- **Load Curves** also accepts `.csv` (`Time` + wells).

4.6 Optional AI skill for input preparation

The GitHub/source repository includes an optional AI-agent skill in the `skills/bioszen-platemap-curves/` [GitHub folder](#). To use it, give that GitHub folder URL to the corresponding AI tool or agent so it can read or acquire the skill. If the agent needs local files, download the repository ZIP from <https://github.com/bioszen/BIOSZEN/archive/refs/heads/main.zip> and copy the `skills/bioszen-platemap-curves/` folder into your agent's skill system.

Use this skill from Codex, Claude, Antigravity, or similar agentic tools when you need to generate a `Datos` + `PlotSettings` platemap from any readable data file, correct an existing platemap, repair parameter-name typing mistakes between `Datos` columns and `PlotSettings$Parameter`, prepare a separate curves workbook, or validate that `Datos$Well` exactly matches curve headers before uploading files to BIOSZEN.

The skill is a documentation/tooling extra. It does not modify the BIOSZEN app and does not assume fixed parameter names, fixed experiment labels, or a specific measurement type.

5. Standard Workflow

1. Load main data file.
2. Optionally load/merge curves.
3. Optionally load metadata.
4. Choose scope (`By Strain` or `Combined`).
5. Select plot type.
6. Apply filters and replicate selections.
7. Optionally normalize by control.
8. Run statistics.
9. Add significance annotations.
10. Export outputs.

Filters and replicates

Exclude replicate(s) in all groups

Filter Media

☒ Select / Deselect all

Media:

- ☒ Ampicillin 1µM
- ☒ Ampicillin 2µM
- ☒ Ampicillin 5µM
- ☒ Control

Replicates by Media

6. Plot Types and Controls

Boxplot

- Best for raw replicate distributions.
- Controls: jitter, box width, point size.
- Supports manual/automatic significance annotations.
- `Flip orientation (horizontal)` improves readability for long group labels.

Barplot

- Best for summarized group comparisons.
- Supports error bars and optional raw points.
- Horizontal orientation available.

Violin

- Best for showing distribution shape with replicate overlays.
- Uses the same annotation workflow as Boxplot/Barplot.
- Horizontal orientation available.

Stacked

- Parameter selector + parameter ordering controls.
- Configurable deviation bars and color behavior.
- Annotation labels/brackets available.
- Horizontal orientation available.

Correlation

- Select X/Y parameters.
- Methods: Pearson, Spearman, Kendall.
- Optional overlays: regression line, `r`, `p`, `R2`, equation.
- Advanced panel supports one-vs-all style screening and Excel export.

Heatmap

- Parameter subset selection.
- Scale options: none, row, column.
- Optional clustering and dendrograms.

- Optional in-cell value labels.

Correlation Matrix

- Multi-select parameters.
- Correlation method + p-value correction.
- Optional significant labels only.

Curves

- Configure axes, labels, line width.
- Choose line geometry and confidence interval style.
- Optional raw replicate trajectories.

Composition Panel

Recommended steps:

1. Click **Add to panel** from individual plots.
2. Open **Composition Panel** tab.
3. Select and order plots.
4. Configure layout (rows/columns, grid, output size).
5. Configure style (legend, fonts, text sizes, palette).
6. Optionally add rich text and per-plot overrides.
7. Export to **PNG** , **PPTX** , **PDF** .

Label (*, **, n.s.):

*

+ Add bar

Update selected label

Remove

Clear all

Current bars

Select bars to reorder/remove

Move up

Move down

Line width:

0.8

Offset:

0.05

Bar spacing:

0.05

Text padding:

0.01

Text size (pts):

5

☐ Hide caps

7. Normalization

Enable **Normalize by control** and pick a control medium.

- BIOSZEN creates normalized columns with `_Norm` suffix.
- Correlation supports axis-specific normalization (`both` , `X only` , `Y only`).
- Fallback logic is applied when strict control pairing is not available.

8. Statistics

Main statistical tools

- Shapiro-Wilk: `stats::shapiro.test`
- Kolmogorov-Smirnov: `stats::ks.test`
- Anderson-Darling: `nortest::ad.test`
- ANOVA: `stats::aov`
- Kruskal-Wallis: `stats::kruskal.test`
- t-test routes: `rstatix::t_test` , `rstatix::pairwise_t_test`
- Wilcoxon routes: `rstatix::wilcox_test`
- Multiple-testing correction: `stats::p.adjust`

Post hoc routes by selection:

- Tukey / Games-Howell: `rstatix`
- Dunn: `rstatix::dunn_test`
- Dunnett: `DescTools::DunnettTest`
- Scheffe, Conover, Nemenyi, DSCF: `PMCMRplus`

Curve statistics (`S1` - `S4`):

- `S1` : `stats::lm` + `splines::ns` + `stats::anova`
- `S2` : `stats::pnorm` + `stats::pchisq`
- `S3` : `stats::pnorm`
- `S4` : `gcplyr::auc` + normality-driven comparisons (`stats::t.test` , `stats::wilcox.test` , `stats::aov` , `stats::kruskal.test`)

Comparison modes:

- All vs all
- Control vs all

- Pair

P-value correction options:

- Holm
- FDR
- Bonferroni
- None

CAUTION: In Summary mode, normality may be **NA** and some non-parametric paths that require raw observations are disabled.

9. Significance Annotations

Manual workflow:

1. Select Group 1 and Group 2.
2. Enter label (***** , ****** , ******* , **ns** , custom text).
3. Add/reorder/edit/remove annotations.

Automatic workflow:

1. Run significance tests.
2. Open auto-annotation options.
3. Choose inclusion (**significant only** or **all**).
4. Choose label mode (**stars** or **p-value**).
5. Replace or append annotations.

10. QC and Replicate Management

Use QC panels to monitor:

- Missing values.
- Outliers by group.
- Sample size and replicate coverage.

Biological replicates

- Manual include/exclude controls.
- Automatic IQR filtering.
- Keep-N reproducibility selection.

Keep-N behavior:

- Ranks replicates by distance to the group median across selected parameters.
- Keeps lowest-score (most reproducible) replicates.

Technical replicates

Available when technical replicates are valid:

- Dedicated technical QC tab.
- Group and biological-replicate selectors.
- Global select/deselect controls.
- Automatic IQR technical outlier detection.
- Technical Keep-N per subgroup.

Replicates by Media

Select all replicates again

Replicates - Control
☒ 1
☒ 2
☒ 3

Replicates - Ampicillin 1µM
☒ 1
☒ 2
☒ 3

Replicates - Ampicillin 2µM
☒ 1
☒ 2
☒ 3

Replicates - Ampicillin 5µM
☒ 1
☒ 2
☒ 3

11. Metadata and Reproducibility

Metadata flow:

- **Download metadata** to save current state.
- Re-import metadata in future sessions.
- Orientation flip state persists across metadata roundtrip.

Reproducibility bundle:

- Save plot versions in-session.
- Export ZIP with plot assets + metadata.
- Reopen workflows with consistent configuration.

Regression coverage includes:

- Orientation flip applied only to Boxplot/Barplot/Violin/Stacked.
- Metadata roundtrip persistence checks.
- Final builder orientation checks.

12. Downloads

Main outputs:

- Plot image (PNG , PDF , depending on chart).
- Data export.
- Metadata export.
- Statistics export.
- Bundle ZIP.
- Advanced correlation tables.
- Merged platemap/curves exports (if merge tools were used).

13. Growth Module

Growth tab file support:

- Accepted file type: Excel (.xlsx).
- Auto-detected structures:
 - Reader/Tecan-like raw layout (typically later-row data in Sheet1).
 - Processed A1 table layout (first column time, following columns as wells/curves).

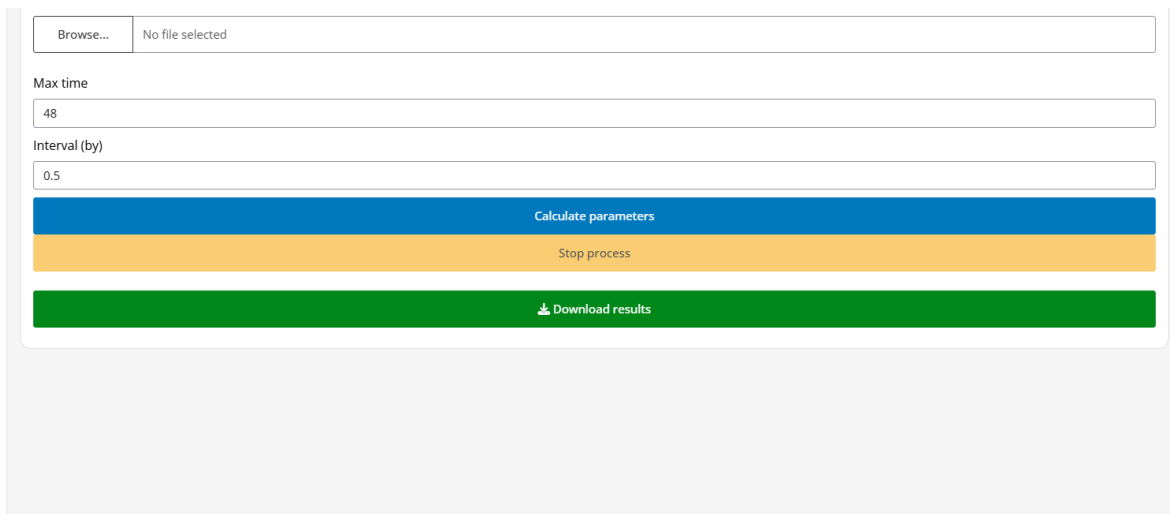
Extracted parameters:

- uMax : maximum exponential-phase slope.
- max_percap_time : time window of max per-capita growth.
- doub_time : doubling time ($\ln(2) / u_{\text{Max}}$).
- lag_time : pre-exponential transition time.
- ODmax : maximum measured OD/signal.
- max_time : time at ODmax .
- AUC : area under the curve.

Typical flow:

1. Upload one or more growth files.

2. Set max time and interval.
3. Run extraction.
4. Download ZIP outputs.
5. Reuse extracted outputs in main plotting workflows.



Browse... No file selected

Max time
48

Interval (by)
0.5

Calculate parameters

Stop process

Download results

14. Troubleshooting Playbook

- **Upload error**
Likely cause: Missing sheet/column names.
What to do: Validate workbook structure and exact headers.
- **No plot generated**
Likely cause: Selected parameter/group absent after filtering.
What to do: Reset filters and verify parameter availability.
- **Only Curves appears in plot type selector**
Likely cause: No valid parameter columns were parsed from the uploaded data file.
What to do: Verify grouped/summary sheet structure and parameter headers, then re-upload.
- **Normalization unavailable**
Likely cause: Missing control medium in current scope.
What to do: Confirm control group exists in active subset.
- **Stats disabled**
Likely cause: Mode/test mismatch.
What to do: Switch test or use a mode with raw-compatible data.
- **Curves merge fails**
Likely cause: Well ID mismatch.
What to do: Match platemap `well` values to curves columns.

- **Grouped/Summary workbook loads but Curves has no data**

Likely cause: Missing embedded curves summary sheet.

What to do: Add `Curves_Summary` (or alias) to the workbook, or upload curves separately in **Load Curves**.

- **CSV not recognized**

Likely cause: Wrong delimiter or missing required headers.

What to do: Check delimiter consistency and required columns.

- **Slow performance**

Likely cause: Too many parameters/overlays at once.

What to do: Reduce active parameters and heavy layers.

15. Support

For support and bug reports: `bioszenf@gmail.com`